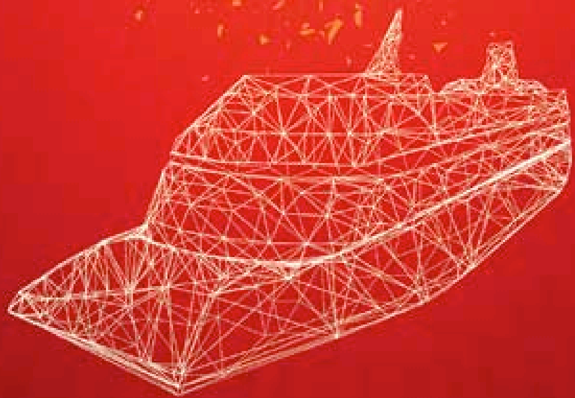
WATER

DRONESHIPS PATROLLING THE SEAS



Droneships are technically known as unmanned surface vehicles or USVs, and the armed ones are known as armed unmanned surface vehicles or AUSVs. Remote controlled USVs were used at the end of the second world war for minesweeping operations, and since then they have been developed for a number of commercial applications. Among the most well known droneships in modern times are those used as remote landing platforms for the reusable SpaceX rockets. There are droneships in operation that function as autonomous ferries, as well as unmanned research vehicles that legitimately conduct long term environmental monitoring and seafloor mapping in the open oceans. The weather ships and oceanographic research vessels that are autonomous can harness the energy of the waves, the wind and sunlight, allowing them to maintain a persistent presence in the oceans for long periods of time. The NOAA has been using “Wave Glider” USVs that harness their energy from the moving waves as well as the sunlight for oceanographic purposes since 2012. This US platform is an Unmanned Underwater Surface Vessel (UUSV) that moves just below the surface with a sail jutting out to capture wind and solar energy. The vehicle can potentially be used for anti mining operations. The company, Ocean Aero has also developed the Triton, an environmentally powered sea vehicle that can fold up its sail to submerge for long term ocean monitoring either at the surface or underwater. Shipping companies are exploring the option of reducing the crews on cargo containers, by making them increasingly autonomous, till full autonomy is achieved. In military applications, uncrewed vessels are expected to provide much the same capabilities as conventional warships, at a fraction of the cost.

There are a number of small autonomous watercraft being developed by the militaries around the world for the purpose of ferrying cargo and crew, deploying and retrieving UUVs, for scouting and extending the range of sensors. The smaller AUSVs in development have capabilities of acting as tugs, operating in formations, and are even mounted with autonomous machine gun turrets, such as the US based Dragon Spy. The Dragon Spy is developed for ISR, weapon delivery, channel mapping and area protection. The larger USVs can be deployed for minesweeping, port surveillance, fleet protection, military cargo delivery, and coastal monitoring. They are of interest to navies that want to protect commercial shipping lanes from pirates in small boats, anti-narcotics and smuggling operations, as well as wide area searches.

Historically, communications in the seas have been a major problem, with naval commanders being given more freedom on how best to carry out their missions as compared to their counterparts on land and in the air. To a great extent, this problem has been mitigated by satellites, however, there is a great advantage of reducing or not using communications in the maritime environment. This is because the potential of detection, interception, jamming or even hostile takeovers are reduced when the deployed machines do not use communications at all, which is why there is a push towards developing more autonomous surface craft. USVs in the future will be capable of waiting or patrolling particular regions for durations of time that have so far not been possible with the use of conventional warships.

A number of countries around the world are developing USVs. The B7 developed by the UAE has a remote video feed for an operator, and is meant for ISR missions, deterring threats, towing sonars, and ocean

Credit: Ares Global

The Turkish AUSV, Ulaq